

CHEHALIS, WA, USA

WMO: 720254

Lat: 46.683N

Lon: 122.983W

Elev: 54

StdP: 100.67

Time Zone: -8.00 (NAP)

Period: 07-19

WBAN: 00119

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.1	-2.9	-8.7	1.8	-3.0	-5.8	2.3	-1.9	8.5	7.1	7.6	7.5	0.5	210	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.0	32.2	20.4	29.8	19.3	27.5	18.4	21.1	29.9	20.0	28.4	19.0	26.3	2.9	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.8	23.5	17.2	12.4	22.6	16.2	11.6	21.4	60.7	30.1	57.4	28.4	53.9	26.6	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	5.8	5.1	-8.7	35.6	2.6	2.5	-10.6	37.4	-12.1	38.9	-13.5	40.3	-15.4	42.1		
(5)			-8.9	23.2	2.4	1.0	-10.6	23.9	-12.1	24.4	-13.4	25.0	-15.2	25.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	5.3	5.8	7.7	9.3	13.3	15.8	18.7	19.1	16.2	11.3	7.4	4.5		
	DBStd	5.91	3.34	3.26	2.76	2.61	3.16	3.08	2.60	2.61	2.75	2.68	3.54	3.65		
	HDD10.0	679	148	121	81	43	6	0	0	0	1	16	92	172		
	HDD18.3	2699	405	351	328	271	157	89	25	22	76	217	329	428		
	CDD10.0	1131	2	3	11	22	109	174	271	282	185	57	13	2		
	CDD18.3	109	0	0	0	0	2	13	37	46	11	0	0	0		
	CDH23.3	1754	0	0	2	9	111	219	596	623	189	4	0	0		
	CDH26.7	638	0	0	0	1	24	78	232	249	53	0	0	0		
(14) Wind	WSAvg	1.8	2.0	2.0	2.1	2.0	1.6	1.5	1.5	1.3	1.4	1.7	2.2	2.1		
(15) Precipitation	PrecAvg	1165	174	128	121	88	60	46	19	32	48	104	179	182		
	PrecMax	1642	372	319	227	195	130	115	60	105	171	236	326	364		
	PrecMin	810	9	4	8	23	9	5	0	0	0	11	32	53		
	PrecStd	193	84	63	48	37	29	24	15	29	35	56	79	75		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	15.9	22.0	24.8	29.1	32.5	35.1	34.8	31.2	23.0	17.5	13.7		
		MCWB	11.0	11.1	13.4	15.5	17.8	21.2	21.1	20.3	20.0	15.6	14.7	12.3		
	2%	DB	12.4	13.0	17.5	20.1	26.4	28.8	32.1	32.2	27.8	19.9	15.0	12.3		
		MCWB	10.7	10.0	11.5	12.5	17.3	19.7	20.0	20.6	18.2	14.7	13.1	11.4		
	5%	DB	11.2	12.3	14.8	17.3	23.7	25.2	29.0	29.1	25.1	17.7	13.7	10.2		
		MCWB	10.2	9.9	10.3	11.2	16.0	17.2	19.6	19.5	17.2	13.8	12.0	9.4		
	10%	DB	10.0	10.9	12.6	14.9	20.9	22.7	27.0	27.2	22.7	16.2	12.5	8.9		
		MCWB	9.2	9.5	9.7	10.2	14.7	16.0	18.4	18.7	16.4	13.5	11.2	8.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.4	12.5	13.9	15.8	19.4	21.9	22.5	22.2	20.8	17.8	15.3	12.9		
		MCDB	13.4	14.1	18.2	23.5	27.4	31.3	32.5	31.7	27.8	21.1	16.6	13.5		
	2%	WB	11.3	11.4	12.3	13.5	17.6	19.8	21.0	20.9	19.2	16.0	13.7	11.3		
		MCDB	11.8	12.7	15.5	18.4	25.5	28.2	29.8	29.9	25.5	18.1	15.0	11.9		
	5%	WB	10.1	10.2	11.3	12.0	16.2	17.9	19.7	20.0	18.0	14.9	12.4	9.5		
		MCDB	10.6	11.6	14.1	15.8	22.7	24.3	28.4	28.4	23.2	16.8	13.4	9.8		
	10%	WB	8.9	9.0	10.2	10.9	14.8	16.6	18.6	18.9	17.1	13.8	11.4	8.1		
		MCDB	9.5	10.2	12.5	13.8	20.0	22.3	25.9	26.2	21.2	15.8	12.3	8.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	7.5	9.3	10.2	12.1	12.1	14.1	14.0	12.4	9.4	7.1	5.7		
		MCDBR	7.5	10.1	13.2	15.4	18.1	17.3	19.5	19.2	17.5	12.3	8.0	6.0		
	5% WB	MCWBR	5.4	6.5	7.5	8.4	9.3	8.5	8.6	8.3	8.6	7.4	5.6	4.6		
		MCWBR	5.9	8.0	11.1	13.2	16.3	16.1	17.8	17.4	14.7	9.7	6.7	5.1		
(40) Clear-Sky Solar Irradiance	taub	taub	0.326	0.315	0.332	0.358	0.378	0.365	0.350	0.354	0.340	0.333	0.326	0.328		
		taud	2.482	2.545	2.509	2.402	2.362	2.422	2.486	2.476	2.522	2.539	2.506	2.424		
	Ebn at Noon	Ebn at Noon	767	867	898	898	888	899	908	893	877	831	761	710		
		Edh at Noon	67	76	90	110	118	112	104	101	88	75	65	64		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.05	1.89	2.79	4.08	5.17	5.67	6.34	5.43	4.04	2.31	1.24	0.87		
	RadStd	RadStd	0.17	0.26	0.36	0.39	0.45	0.57	0.48	0.40	0.37	0.27	0.16	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+10

Nomenclature: See separate page